

# MIARNET: MULTI-SCALE INTERACTION AND ADAPTIVE WEIGHTING FOR REFERRING IMAGE SEGMENTATION IN LOW-ALTITUDE UAV IMAGERY

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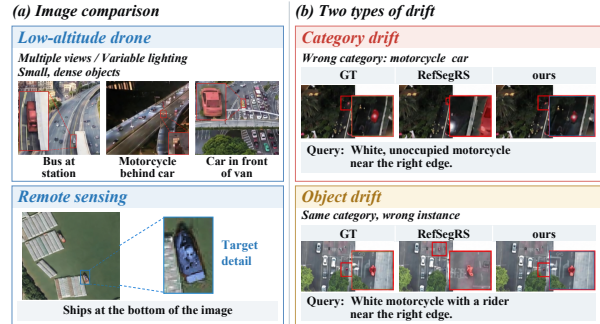
## ABSTRACT

Referring image segmentation (RIS) identifies and segments image targets specified by natural-language descriptions. Although recent methods have advanced conventional and high-altitude remote sensing scenarios, low-altitude UAV imagery remains challenging because diverse viewpoints, densely packed tiny objects, and substantial illumination variations are poorly represented in existing ground-level or fixed-overhead datasets. Existing approaches also inadequately capture global context and local details, leading to category drift, while insufficiently adaptive feature fusion preserves redundant information, reducing discriminability and causing object drift. To address these issues, we introduce DroneRIS, a fine-grained benchmark comprising 10,851 annotated triplets collected from real-world UAV videos, with an emphasis on dense small objects and diverse viewpoints. We further propose a segmentation network equipped with multi-scale feature interaction and extraction (MFIE) and adaptive feature weighting (AFW). MFIE employs interacting branches to jointly model global context and local details, whereas AFW dynamically reweights and fuses MFIE’s input and output features to suppress redundancy and enhance discriminability. Our model outperforms most competitive methods on the primary oIoU metric. Ablation studies verify each module’s effectiveness, yielding consistent gains on P@0.5, P@0.7, and P@0.9. Datasets and Code are available at <https://github.com/lumenworks0403/MIARNET>.

**Index Terms**— Referring image segmentation; Low-altitude UAV imagery; Multi-scale feature interaction; Adaptive feature weighting; DroneRIS benchmark.

## 1. INTRODUCTION

Referring image segmentation (RIS) aims to localize and segment specific image regions according to textual descriptions [1]. Unlike semantic and instance segmentation, it focuses on the target specified by the description rather than all regions belonging to a particular category. In practical applications, RIS enables rapid target localization and precise segmentation based on appearance descriptions, reducing manual effort while offering broad application potential and substantial research value. Multi-scale RIS methods



**Fig. 1.** (a) UAV versus remote sensing imagery. (b) Category and object drift in referring image segmentation. jointly capture global semantic information and local details to enhance robustness in complex scenes [2]. Yu et al. matched appearance, location, and relationship modules with hierarchical visual features to achieve cross-scale semantic alignment. Huang et al. progressively constructed spatial graphs from different word types to model relationships among multi-level features. Other studies improved small-object segmentation efficiency through positional prior embedding and sequential boundary prediction. However, tiny objects and complex backgrounds in low-altitude UAV imagery hinder the integration of global context and local details, potentially shifting predictions toward objects of a different semantic category. We term this failure category drift, as shown in Fig. 1(b).

Adaptive fusion improves target discrimination by selecting informative cues. Dai et al. assigned complementary weights to local and global feature streams, while Yang et al [3] introduced language gating to regulate multimodal information integration and strengthen semantic awareness of the target. However, coupled weights or explicit regulation of only one stream constrain the contributions of individual features. Inadequate preservation of target-specific cues and suppression of distracting responses during fusion may cause the model to select another instance of the same category instead of the referred target. We refer to this failure as object drift, also illustrated in Fig. 1(b).

Remote sensing RIS datasets provide pixel-level annotations and natural-language referring expressions [4, 5, 6], supporting representative studies that demonstrate feasibility and establish standardized evaluation benchmarks (Shi and Zhang 2025; Chen et al. 2025; Pan et al. 2024). Nevertheless, their satellite or helicopter imagery typically features rela-

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tively fixed viewpoints, larger objects, and limited variation in illumination and background. Fig. 1(a) highlights the diverse viewpoints, varying illumination, and dense small objects in low-altitude UAV imagery.

To address these challenges, we first introduce DroneRIS, a fine-grained benchmark for low-altitude UAV referring segmentation, featuring dense small objects, diverse viewpoints, and complex backgrounds. It enables standardized evaluation of tiny-object discrimination, fine-grained text–target alignment, and precise segmentation. Building on this benchmark, we propose MIAWNet, which combines global–local interaction with adaptive weighting to integrate complementary contextual and detail information while suppressing redundant features. Experiments show that MIAWNet outperforms most state-of-the-art methods on the primary oIoU metric.

## 2. DRONERIS DATASET

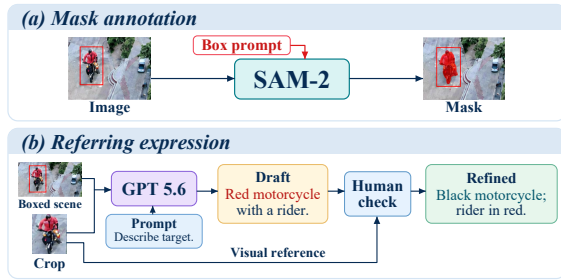


Fig. 2. Semi-automatic annotation pipeline for DroneRIS.

DroneRIS is a fine-grained RIS benchmark designed for low-altitude UAV imagery. Images from CO-Drone [7] cover urban scenes under varied viewpoints and lighting, cropped into  $1080 \times 1080$  patches and manually screened. It includes eight categories: people, car, motor, bicycle, tricycle, truck, bus, and boat. Over 90% of instances have mask coverage below 10%, emphasizing small, densely distributed objects.

Fig. 2 illustrates the annotation pipeline. SAM-2 [8] generates instance masks using oriented bounding boxes from CO-Drone. GPT 5.6 uses cropped instances and full images marked with red bounding boxes to generate detailed appearance and spatial descriptions. Both masks and expressions are manually verified and refined. The 10,851 triplets are split into training, validation, and test sets at approximately 7:1:2, with video sequences kept disjoint across splits to prevent scene leakage. As summarized in Table 1, compared with existing RRSIS datasets [4, 5, 6], DroneRIS offers viewing angles ranging from  $30^\circ$  to  $90^\circ$ , substantial viewpoint variation, and nighttime scenes, supporting RIS evaluation in complex low-altitude scenarios.

## 3. METHOD

Fig. 3 shows the overall architecture. Swin Transformer [9] and BERT [10] extract visual and textual features, which cross-modal attention aligns spatially. The Multi-scale Feature Interaction and Extraction Module combines global context and local details through branch interaction. The Adaptive Feature Weighting Mechanism balances pre- and post-interaction features to suppress redundancy and enhance discriminability. Fused features are concatenated with cor-

Table 1. Comparison of DroneRIS and selected existing referring remote sensing image segmentation datasets.

| Dataset                | Image Source             | Shooting Angle                                     | Nighttime Scene |
|------------------------|--------------------------|----------------------------------------------------|-----------------|
| RefDIOR                | Google Earth             | fixed                                              | ✗               |
| NWPU-Refer             | Google Earth             | fixed                                              | ✗               |
| RISBench               | Google Earth, GF-2, JL-1 | fixed                                              | ✗               |
| RefSegRS               | Helicopter (> 1000 m)    | fixed                                              | ✗               |
| RRSIS-D                | Google Earth             | fixed                                              | ✗               |
| <b>DroneRIS (ours)</b> | <b>Drone (CO-Drone)</b>  | <b><math>30^\circ</math>–<math>90^\circ</math></b> | <b>✓</b>        |

responding visual features, aggregated across stages, and decoded and upsampled to produce a refined segmentation mask at the input resolution.

### 3.1. Multi-scale Feature Interaction and Extraction

The MFIE module combines global attention and local convolutions to capture complementary contextual and spatial information. Given  $Q_x$ ,  $K_x$ , and  $V_x$  projected from  $G_i$ , the global branch attends to pooled keys and values:

$$X_{\text{global}} = \text{Softmax} \left( \frac{Q_x [\text{Pool}(K_x)]^T}{\sqrt{d}} \right) \text{Pool}(V_x). \quad (1)$$

The local branch extracts spatial and channel information through convolutions:

$$\begin{aligned} Q_i &= \text{DWconv}(Q_x; \Theta_q), \\ K_i &= \text{DWconv}(K_x; \Theta_k), \\ V_i &= \text{DWconv}(V_x; \Theta_v), \end{aligned} \quad (2)$$

Element-wise query–key products undergo linear transformations and nonlinear activations to generate context-aware weights, which reweight value features to enhance details and boundaries:

$$X_{\text{local}} = \text{Tanh} \left( \frac{W_2 \text{Swish}(W_1(Q_i \odot K_i))}{\sqrt{d}} \right) \odot V_i. \quad (3)$$

Finally, concatenation and joint projection fuse global context and local details:

$$G'_i = W_o \text{Concat}(X_{\text{local}}, X_{\text{global}}). \quad (4)$$

Here, Pool denotes average pooling, DWconv denotes depthwise convolution, and  $\odot$  denotes element-wise multiplication.  $d$  is the channel dimension; and  $\Theta$  and  $W$  are learnable parameters.

### 3.2. Adaptive Feature Weighting Mechanism

To reduce redundancy during feature fusion and enhance target discriminability, we introduce an adaptive feature weighting mechanism. As shown in Fig. 3, it takes the input features  $G_i$  and the enhanced features  $G'_i$  from the Multi-scale Feature Interaction and Extraction Module, generating content-dependent weights to regulate their contributions.

Specifically,  $G'_i$  passes through a  $1 \times 1$  convolution and GELU activation for channel mixing and nonlinear transformation, followed by another  $1 \times 1$  convolution and Tanh activation to generate weights  $S_i$ . The same processing structure

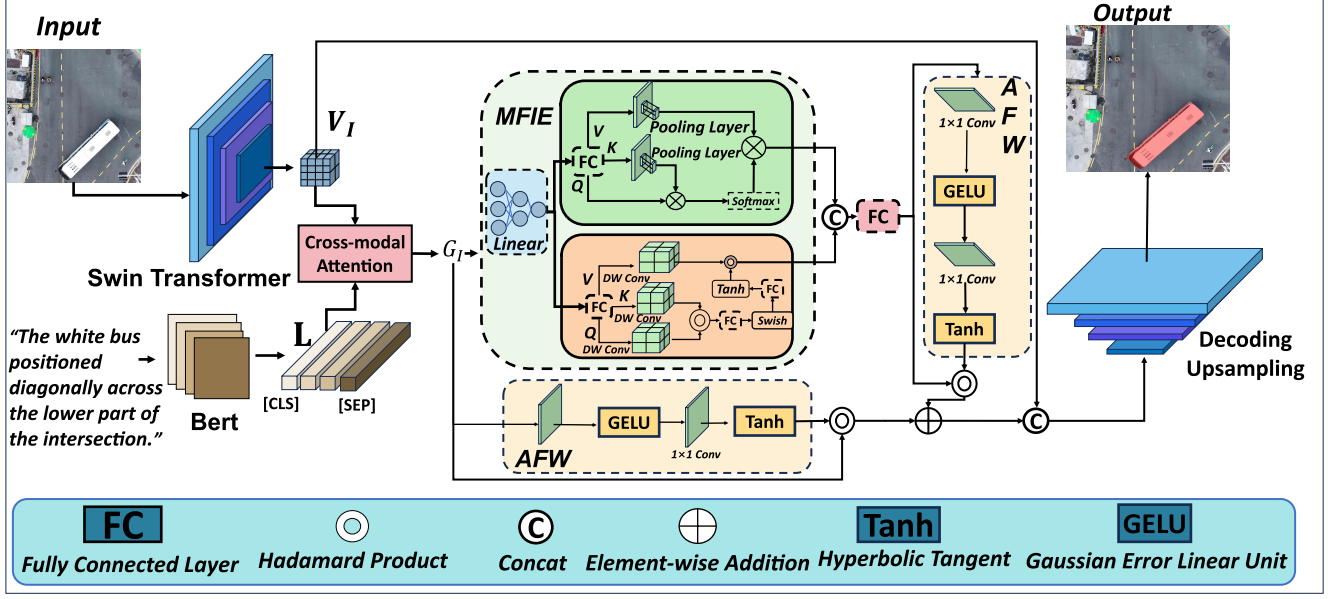


Fig. 3. Overall architecture of the proposed method.

generates weights  $Y_i$  from  $G_i$ . Each stream thus derives its weights from its own features:

$$S_i = \varphi_i(G'_i), \quad (5)$$

$$Y_i = \gamma_i(G_i), \quad (6)$$

where  $\varphi_i$  and  $\gamma_i$  denote weighting mappings comprising two  $1 \times 1$  convolutions with GELU and Tanh activations. Each feature stream is multiplied element-wise by its corresponding weights, and the results are summed to obtain fused semantic features  $F$ :

$$F = S_i \odot G'_i + Y_i \odot G_i, \quad (7)$$

where  $\odot$  denotes element-wise multiplication. This lightweight gating mechanism adaptively regulates both streams without fixed fusion ratios or complementary-weight constraints. Content-dependent weights allow the model to exploit informative cues from both the original and multi-scale interaction features. In low-altitude UAV scenes, the mechanism preserves subtle target cues while suppressing background responses and redundancy, improving feature discriminability for precise segmentation.

### 3.3. Loss Function

To address small targets and foreground-background pixel imbalance in UAV imagery, we combine binary cross-entropy (BCE) loss with Dice loss to supervise the final segmentation mask. The two terms provide complementary constraints on pixel-wise classification and region-level overlap.

Let  $p_j \in (0, 1)$  denote the predicted foreground probability at pixel  $j$ , and  $m_j \in \{0, 1\}$  the corresponding ground-truth label. The BCE loss is defined as:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{j=1}^N [m_j \log p_j + (1 - m_j) \log(1 - p_j)], \quad (8)$$

where  $N$  is the number of pixels in the output mask. BCE provides direct supervision for distinguishing target pixels from the background. However, the large proportion of background pixels in small-target scenes can dominate pixel-wise supervision.

We therefore introduce Dice loss to encourage overlap between the predicted and ground-truth regions and help mitigate foreground-background imbalance:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{j=1}^N p_j m_j + \epsilon}{\sum_{j=1}^N p_j + \sum_{j=1}^N m_j + \epsilon}, \quad (9)$$

where  $\epsilon$  is a small constant for numerical stability. Continuous prediction probabilities are used to preserve differentiability.

The overall training objective combines the two losses with equal weights:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}. \quad (10)$$

The loss is computed per sample and averaged over the batch. Supervision on the final mask jointly optimizes the feature interaction, adaptive weighting, and decoding modules without additional auxiliary losses.

## 4. EXPERIMENTS

### 4.1. Experimental Setup

We evaluate on RefCOCO [22], RefCOCO+, and DroneRIS, using images from MS COCO [23] for the former two and CO-Drone [7] for the latter. RefCOCO includes spatial descriptions, while RefCOCO+ excludes absolute location words and emphasizes appearance. DroneRIS provides longer and more fine-grained referring expressions. Evaluation uses the validation, TestA, and TestB splits of RefCOCO and RefCOCO+, and DroneRIS validation. Metrics are oIoU (aggregated intersection divided by aggregated union), mIoU (mean per-sample IoU), and P@X (percentage of samples

**Table 2.** Comparison with existing methods on three benchmarks in oIoU (%).

| Method        | Backbone          | RefCOCO      |              |              | RefCOCO+     |              |              | DroneRIS     |
|---------------|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|               |                   | Val          | TestA        | TestB        | Val          | TestA        | TestB        | Test         |
| EFN [11]      | ResNet-101        | 62.76        | 65.69        | 59.67        | 51.50        | 55.24        | 43.01        | 51.13        |
| BUSNet [12]   | ResNet-101        | 63.27        | 66.41        | 61.39        | 51.76        | 56.87        | 44.13        | 51.79        |
| CGAN [13]     | DarkNet-53        | 64.86        | 68.04        | 62.07        | 51.03        | 55.51        | 44.06        | 51.01        |
| VLT [14]      | DarkNet-53        | 65.65        | 68.29        | 62.73        | 55.50        | 59.20        | 49.36        | 52.99        |
| SeqTR [15]    | DarkNet-53        | 67.26        | 69.79        | 64.12        | 54.14        | 58.93        | 48.19        | 59.68        |
| MaIL [16]     | ViT-B-16          | 69.38        | 71.31        | 66.76        | 62.23        | 65.92        | 52.06        | 62.45        |
| CRIS [17]     | ResNet-101        | 70.47        | 73.18        | 66.10        | 62.27        | 68.08        | 53.68        | 59.87        |
| ETRIS [18]    | ViT-B-16          | 70.51        | 73.51        | 66.63        | 60.10        | 66.89        | 50.17        | 59.82        |
| CNNFormer [2] | ResNet-50+SwinT-B | 71.93        | 76.13        | 67.74        | 61.78        | 67.78        | 51.77        | 60.84        |
| BarLeRIa [19] | CLIP-B            | 72.40        | 75.90        | 68.30        | 64.00        | 69.80        | 55.50        | 62.40        |
| LAVT [3]      | SwinT-B           | 72.73        | 75.82        | 68.79        | 62.14        | 68.38        | 55.10        | 61.24        |
| MagNet [20]   | SwinT-B           | 74.24        | 77.24        | 70.15        | 65.16        | 70.32        | 57.14        | 64.36        |
| RISCLIP [21]  | CLIP-B            | 73.57        | 76.46        | 69.76        | 64.53        | 69.61        | 55.09        | 63.10        |
| Ours          | SwinT-B           | <b>75.96</b> | <b>78.09</b> | <b>70.26</b> | <b>66.31</b> | <b>71.12</b> | <b>58.96</b> | <b>73.59</b> |

with IoU exceeding  $X$ ), where  $X \in \{0.5, 0.7, 0.9\}$ .

Experiments use Python 3.10.0 and PyTorch 2.0.0 on Ubuntu 22.04 with an NVIDIA RTX 4090. Visual and text encoders are ImageNet-1K-pretrained Swin Transformer-Base and BERT, respectively [24, 9, 10]. Images are resized to  $480 \times 480$  and normalized without additional augmentation. Training runs for 50 epochs using AdamW [25] with batch size 8, weight decay 0.05,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ , and  $\epsilon = 10^{-8}$ . The learning rate follows cosine decay from  $3 \times 10^{-5}$  to  $1 \times 10^{-6}$ . The checkpoint with the highest validation mIoU is evaluated on the corresponding test split, where available. All competing methods on DroneRIS are retrained using the same data splits and their official implementations.

#### 4.2. Comparison with Existing Methods

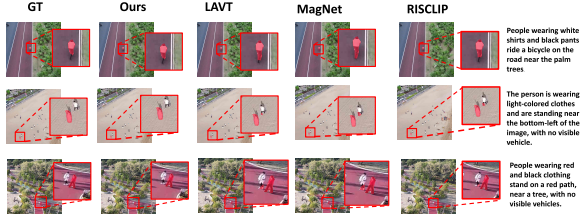
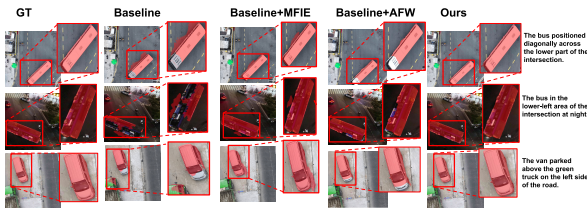
**Fig. 4.** Qualitative comparison with existing methods.

Table 2 compares our method with existing approaches on RefCOCO, RefCOCO+, and DroneRIS, showing improvements over the baseline on all three datasets. On RefCOCO, our method achieves 78.09% and 70.26% oIoU on TestA and TestB, respectively, demonstrating effective person and object segmentation and suggesting improved text–visual alignment through local–global modeling.

**Fig. 5.** Qualitative ablation results on DroneRIS.

On RefCOCO+, our method demonstrates strong attribute understanding by leveraging appearance and context without explicit location words. Performance on TestB remains lower than TestA, consistent with the greater difficulty of non-person referring cases. On the DroneRIS test set, our method achieves 73.59% oIoU, demonstrating its effectiveness for low-altitude UAV referring segmentation. Fig. 4 provides qualitative comparisons. MagNet [20] introduces an additional mask encoder and ground-truth masks during training, increasing supervision requirements and model complexity. In contrast, our method achieves competitive performance without additional mask supervision.

#### 4.3. Ablation Study

We conduct ablations on the DroneRIS validation split by progressively adding the feature interaction and extraction module and adaptive feature weighting to our SwinT-B/BERT cross-modal baseline, keeping the data split, backbone, and training settings fixed. Table 3 shows that introducing the interaction module increases mIoU from 74.46% to 75.30% and oIoU from 72.73% to 73.12%. Adaptive feature weighting further improves these metrics to 75.88% and 73.59%, yielding total gains of 1.42 and 0.86 percentage points, respectively. These results support the effectiveness of both components. Fig. 5 compares the qualitative results of the baseline, baseline+MFIE, baseline+AFW, and the full model.

**Table 3.** Ablation results on DroneRIS.

| MFIE | AFW | mIoU (%)     | oIoU (%)     | P@0.5        | P@0.7        | P@0.9        |
|------|-----|--------------|--------------|--------------|--------------|--------------|
| —    | —   | 74.46        | 72.73        | 74.44        | 65.58        | 30.23        |
| ✓    | —   | 75.30        | 73.12        | 74.95        | 66.06        | 30.30        |
| —    | ✓   | 74.92        | 73.04        | 75.63        | 66.84        | 30.48        |
| ✓    | ✓   | <b>75.88</b> | <b>73.59</b> | <b>76.97</b> | <b>68.32</b> | <b>31.02</b> |

## 5. CONCLUSION

We introduced DroneRIS, a benchmark with 10,851 image-text-mask triplets for low-altitude UAV referring segmentation, and a network with multi-scale feature interaction and adaptive weighting. Experiments on three datasets show consistent improvements, with the two modules jointly increasing mIoU and oIoU by 1.42 and 0.86 percentage points, respectively. Future work will focus on challenging scenes, cross-scene generalization, and efficiency.

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